

DM-UY 3193

Dynamic Web Applications

Fall 2026 • Professor Kathryn Adee

SECTION A Tuesday & Thursday, 8:00–9:50 AM 370 Jay Street, Room 307 • Brooklyn Campus	SECTION B Tuesday & Thursday, 10:00–11:50 AM 370 Jay Street, Room 307 • Brooklyn Campus
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Instructor: Kathryn Adee • kaa328@nyu.edu

Office hours: [viewable](#) and [bookable](#) online

Course Prerequisites

Introduction to Front End Web Development, or equivalent knowledge. Moderate familiarity with HTML5, CSS3, and JavaScript is expected.

Course Description

This course is a studio in modern client-side web application development. Students build single-page applications with React, working from individual components up to a complete, data-driven application of professional quality in both design and production. Assignments are arranged in sequence, and the studio stresses interactivity, usability, and the quality and appropriateness of look and feel, using the design and build patterns current in professional front-end practice.

We begin with React fundamentals — JSX, reusable components, props, and state — and with the discipline of breaking an interface into small, composable pieces. From there students build a component library with scoped and utility-first styling, learning to design components intended to be reused by other developers rather than written once and abandoned.

The second half moves from components to applications: client-side routing, fetching and rendering data from external APIs, and managing state once it outgrows a single component, using React's Context API and Redux Toolkit. The semester closes with an introduction to type safety in TypeScript and a self-directed final project.

Program Learning Objectives

Students will:

- develop conceptual thinking skills to generate ideas and content in order to solve problems or create opportunities;
- develop technical skills to realize their ideas;

- develop critical thinking skills that will allow them to analyze and position their work within cultural, historic, aesthetic, economic, and technological contexts;
- gain knowledge of professional practices and organizations by developing their verbal, visual, and written communication for documentation and presentation, exhibition and promotion, networking, and career preparation;
- develop collaboration skills to actively and effectively work in a team or group.

Course Objectives

- Refresh and expand knowledge of the fundamental front-end languages — HTML5, CSS3, and modern JavaScript.
- Create and build upon a React application using current professional tooling, including a modern build tool, npm, and Git.
- Compose interfaces from reusable components, and reason clearly about props, state, and where state belongs.
- Track user interaction with React state, the Context API, and Redux Toolkit.
- Build and maintain a component library using CSS Modules, CSS custom properties, and utility-first CSS.
- Fetch, render, and manage data from external APIs.
- Apply type safety through an introduction to TypeScript.
- Document all coursework in a public GitHub repository using Markdown.

Course Structure

Class time is spent as a combination of lecture, review, live coding, and in class exercises and presentations. Homework consists of online tutorials as well as weekly projects that build on each week's topics of discussion.

It is expected that everyone in the class will create and maintain a repository with README .md files for documentation.

Readings

There is no required textbook for this course. Students are required to review the upcoming class notes, which are published in the class repository.

Laptop Usage During Class

I love having WiFi available and I think it is a valuable tool for use in the classroom. Unfortunately, it can very easily be a distraction as well. We are going to be exploring a variety of models for live interaction online and therefore laptops will prove very useful. We should though be courteous to one another, and when we have outside guests, during discussion, or during student presentations, laptop use should be curtailed.

Grading

COMPONENT	WEIGHT
Regular Assignments	20%
Weekly assignments relevant to material from the previous class.	
Mid-Term Project	25%
A project demonstrating working knowledge of React components, props, and state.	
Participation and Attendance	25%
Attendance is mandatory. This class is participatory: you are expected to take part in discussions and to give feedback on other students' projects.	
Final Project	25%
Both technology and content driven, built over the course of several weeks.	
Documentation	5%
IDM coursework documentation, kept in your GitHub repository for this class using Markdown files.	
Total	100%

Attendance. Please inform me by email if you are going to miss a class. Arriving late, or an excessive number of absences, will adversely affect your grade. **Three unexcused absences will result in your grade dropping by 5%.**

Qualitative Grading Overview

Each student will be judged on the quality, experimentation, and improvement that their work shows.

A 90-100	Performance, participation, and attendance have been of the highest level, showing sustained excellence in meeting course responsibilities. Work clearly differentiates itself, has memorable impact, and pursues concepts and techniques above and beyond what is discussed in class. The student thoroughly understands the web design and development process.
B 80-89	Performance, participation, and attendance have been good, though not of the highest level. Work demonstrates a better than average web design and development process.
C 70-79	Performance and attendance have been adequate, satisfactorily meeting the course requirements. Work is average and competent, showing a basic understanding of the web design and development process.
D 60-69	Performance and attendance have been less than adequate. Work is lacking in many or most areas that show any understanding of visual foundation. Problems may include lack of interest, procrastination, poor planning, and poor craft.

F 59 & below	Performance and attendance have been such that course requirements have not been met. Work shows no overall understanding of the course material on many levels, or a severe lack of interest.
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Accommodations

If you are a student with a disability who is requesting accommodations, please contact New York University's **Moses Center for Student Accessibility (CSA)** at 212-998-4980 or mosescsa@nyu.edu. You must be registered with CSA to receive accommodations. Information about the Moses Center can be found at nyu.edu/csa. The Moses Center is located at 726 Broadway, 3rd Floor.

Notable Dates

Add / Drop deadlines

After these dates students may not add classes, and dropped classes will result in a grade of W.

- First 7-week session classes — **Tuesday, September 8**
- Full semester classes — **Tuesday, September 15**
- Second 7-week session classes — **Wednesday, October 28**

Pass/fail request and withdrawal deadlines

Deadline to request the pass/fail grading option, where school policy allows, and to withdraw from classes.

- First 7-week session classes — **Friday, October 9**
- Full semester classes — **Wednesday, November 25**
- Second 7-week session classes — **Wednesday, November 25**

Classes will not meet

- Labor Day — **Monday, September 7**
- Fall Break — **Monday, October 12**
- Thanksgiving Recess — **Thursday, November 26 - Friday, November 27**

Legislative (make-up) day

- **Wednesday, October 14** — Monday classes will meet instead of Wednesday classes.

Registration

- January 2027 registration opens — **Tuesday, October 13**
- Spring 2027 registration appointments begin — **Monday, November 16**

End of semester

- Incomplete grade request deadline, undergraduate students only — **Tuesday, December 22**
- Reading Day — **Tuesday, December 15**. Reading days allow students the maximum opportunity to prepare for final examinations; final examinations and required meetings or classes are not scheduled on a reading day.

This course meets Tuesdays and Thursdays. Of the dates above, only the Thanksgiving Recess on Thursday, November 26 falls on a scheduled class day. These and other important deadlines can be viewed on the [academic calendar published by the Office of the Registrar](#), which may also be added to a personal Google calendar.